Bikash Kunwar

AUBURN, AL • (571) 530-7381 • BZK0067@AUBURN.EDU

Flight-Test Media (2022–Present): https://www.vsddl.com/flight-testing/
Projects (2014–2022): https://b-kunwar.github.io/

EDUCATION

Ph.D. in Aerospace Engineering *(Aug. 2022 – Present)*
Auburn University, AL, USA

B.E. in Mechanical Engineering *(Nov. 2015 – Nov. 2019)*
Tribhuvan University, Pulchowk Campus, Lalitpur, Nepal

- Final-year project – Design, Fabrication, and Test of Unmanned Aerial Vehicle

RESEARCH EXPERIENCE

Graduate Research Assistant *(Aug. 2022 – Present)*
Vehicle Systems, Dynamics, and Design Laboratory, Auburn University, AL, USA

- **Aircraft Model Development:** Support rapid full-scale and subscale aircraft sizing and simulation for Advanced Air Mobility (AAM) applications.
- **Flight Testing and System Identification:** Perform experimental characterization of subscale aircraft for Simplified Vehicle Operation (SVO)–centric AAM applications.
- **Simulation Model Calibration and Validation:** Use flight-test data to calibrate and validate simulation models.
- **In-Flight Control Effector Fault Detection and Controller Reconfiguration:** Develop solutions to enable constrained flight-envelope protection after effector failure.

FLIGHT TEST ENGINEERING EXPERIENCE

Graduate Research Assistant *(Aug. 2022 – Present)*
Vehicle Systems, Dynamics, and Design Laboratory, Auburn University, AL, USA

- **Aircraft Supported:** Lift-plus-cruise VTOL (LPC-03-s Phoenix Lite), tilt-wing VTOL (Vy 5.3.1 RRTV, VT-04-s Albatross), multi-tilt-rotor CTOL/VTOL (VT-03-s Shadow, VT-05-s Griffin), fixed-wing (RN-001-AE-s), and blended-wing-body (LV001-s).
- **Flight Test Planning:** Participate in flight-test planning and rehearsals; plan and execute system identification and vibration tests.
- **Flight Readiness and Preflight Checks:** Evaluate pilot-in-the-loop and automated missions using a dual-purpose GCS/simulator; verify sensors, propulsion, and GCS–aircraft communication.
- **Flight Briefings and Debriefings:** Brief simulation and system-check results; review flight observations, telemetry displays, and preliminary data plots.
- **Experimental Flight Testing:** Serve as a GCS flight test engineer, monitoring instrumentation, identifying anomalies, and communicating findings to the flight-test team.

- **Post-Flight Analysis:** Analyze telemetry in MATLAB; generate flight-data plots and telemetry/video playback for performance evaluation.
- **Simulation and Control Development:** Develop and simulate subscale aircraft models; develop and test basic control laws (ground steering, RPM control).
- **Hardware and Communication Integration:** Test and integrate motors, ESCs, batteries, and companion computers; implement DroneCAN and MAVLink communication.
- **Flight Test Documentation:** Prepare preflight checklists, flight-test procedures, post-flight summaries, system architecture descriptions, and supporting technical documentation.

Mechanical Design Engineer for Drones *(Jan. 2020 – May 2022)*
Prokura Innovations, Lalitpur, Nepal

- **Aircraft Design, Development, and Flight Testing:** Designed and developed lift-plus-cruise VTOL UAVs for medicine delivery with UNICEF Innovation Fund support.

TECHNICAL SKILLS

Microsoft Office (Word, Excel, PowerPoint), Outlook, MATLAB/Simulink, PX4, QGroundControl, DroneCAN, MAVLink, Flight Data Analysis, System Identification

COMPETITION PARTICIPATION

Design-Build-Vertical-Flight (DBVF, 2023–2024), GoAERO (2025-Present)

HONORS AND AWARDS

- **AIAA Orville and Wilbur Wright Graduate Award (2026)**
- **GoAERO Prize: Stage 1 NASA U.S. University Innovation Project Winner and Stage 2 Winner (2025)** — Awarded to Team VSDDL, Auburn University**.**

SELECTED PUBLICATIONS

- **Kunwar, B.**, Putra, S. H., Bhandari, R., McCormick, H. C., and Chakraborty, I., "Simulation Model Calibration Using System Identification for a Vectored Thrust Multi-Tilt-Rotor Configuration," ICAS 2026, Paper 929, Sydney, Australia, September 2026.
- Chakraborty, I., **Kunwar, B.**, Putra, S. H., Bhandari, R., and McCormick, H. C., "Flight Testing of the VT-03-s Shadow Multi-Tilt-Rotor Configuration in VTOL and CTOL Operations," AIAA SciTech 2026 Forum, January 2026. https://doi.org/10.2514/6.2026-2084

TEACHING AND MENTORING EXPERIENCE

Graduate Teaching Assistant (*Aug. 2024 – Present)*
Auburn University, AL, USA

- **Aero 3230 Flight Dynamics** *(Fall 2024, Summer 2025, Fall 2025, Spring 2026)*
- **Aero 3220 Aerospace Systems** *(Spring 2025)*

LEADERSHIP AND SERVICE

President *(Aug. 2024 – Aug. 2025)*
Nepalese Students Association (NSA), Auburn University